AI is the future. Everyone should be able to use it, even with no internet.

Registered partnership firm, Andhra Pradesh | Applying for Startup India (DPIIT) recognition



THE PROBLEM

Every job will use AI. You learn it by using it. Not everyone in India can.

Human-guided AI: AI is becoming part of every job, with people guiding it. People learn it by doing.

Cloud barriers: Cloud AI needs reliable internet and charges for every query.

Connectivity gap: Over a third of Indian schools have no internet (UDISE+ 2024-25: 63.5% connected).

Farm scale: About 86% of Indian farm holdings are small or marginal (Agriculture Census 2015-16).

Connected city phone
High bandwidth, costly cloud

Always on

Bridged by the PocketMind engine



Village phone
No signal, no cloud budget

Offline first

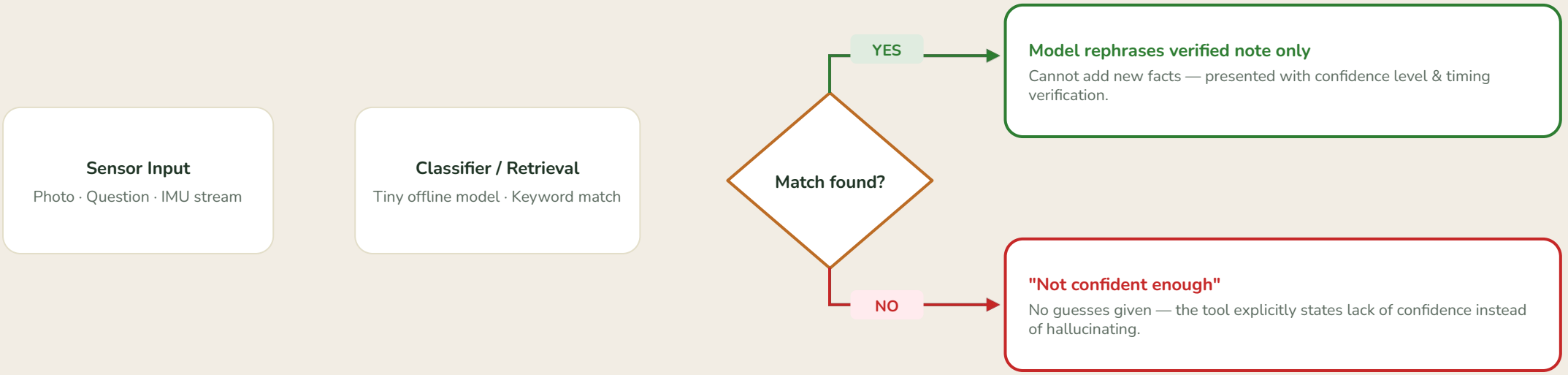
HOW IT WORKS

Grounded by design, not by hope.

LANGUAGE LAYER

Speak or type in any 9* Indian language → Detect language → Translate in → Small model answers → Translate back

GROUNDING ENGINE & VERIFICATION FLOW



* The 9 languages are English, Bengali, Gujarati, Hindi, Kannada, Malayalam, Marathi, Tamil, and Telugu - and 10 more languages in progress.

ENGINEERING METHOD

Spend compute once in the cloud. Then run intelligence for free on the device.

STEP 1
Define the task

STEP 2
Teach with large model

STEP 3
Adapt open small model

STEP 4
Compress

STEP 5
Test rigorously

STEP 6
Ship and update

Near-zero marginal cost

Cost per extra user is close to zero because the computing happens on the user's own device.

Seamless updates

Models update whenever a connection appears, so the device keeps getting better.

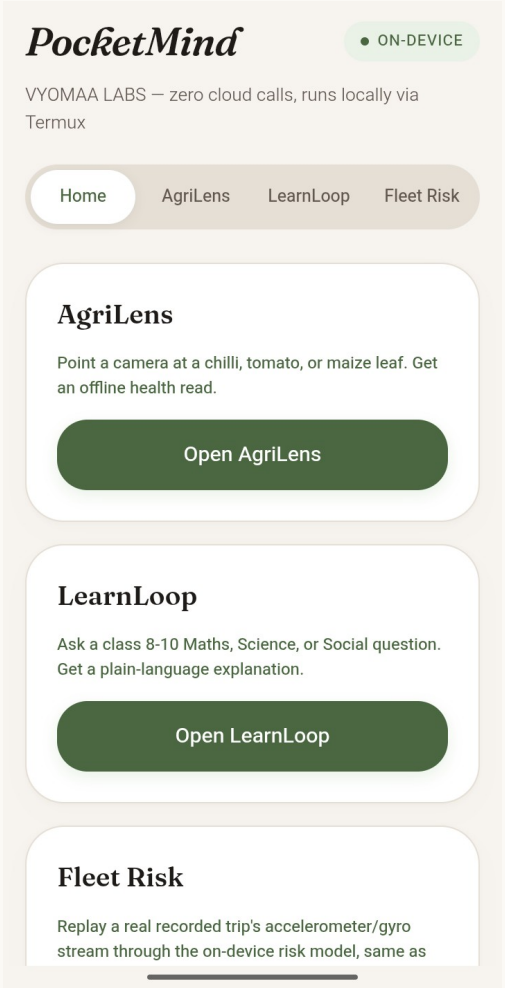
THE PLATFORM

Build the engine once. Serve farming, learning, health, environment and vehicles.

AgriLens
Crop health
Now

LearnLoop
AI tutor
Now

Edge fleet unit
Vehicle safety
Now



MedNotes
Clinical records
Next

EcoTrace
Biodiversity map
Next

FIRST PRODUCTS

From a crop photo to a patient tutor, running offline.

AgriLens

Point the camera at a crop and get an instant health read in a small app that needs no signal. Advice comes from a verified local advice bank.

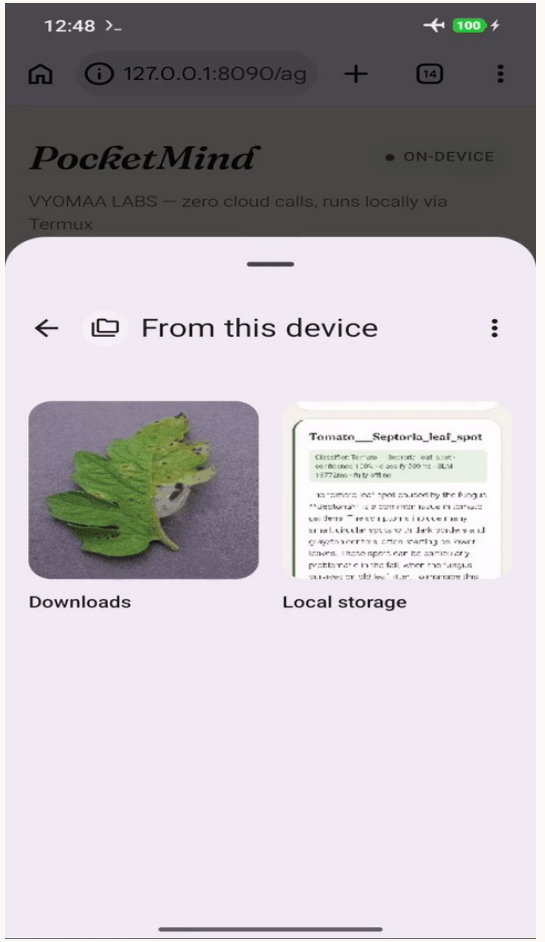
Stage: Development and internal testing phase is freezed and Field testing with 5 Farmers of different crops.

LearnLoop

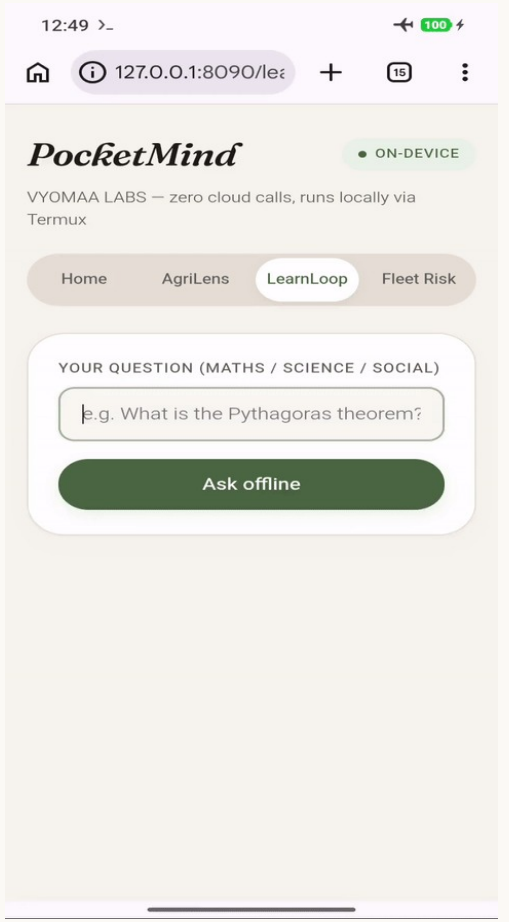
A patient practice partner for students to learn and think aloud in their own language, on the device they already have.

Stage: Phase 1 Development, battle testing is completed, Field testing with 10 rural students in addition to teaching AI technology for 5 hours a week

AgriLens



LearnLoop



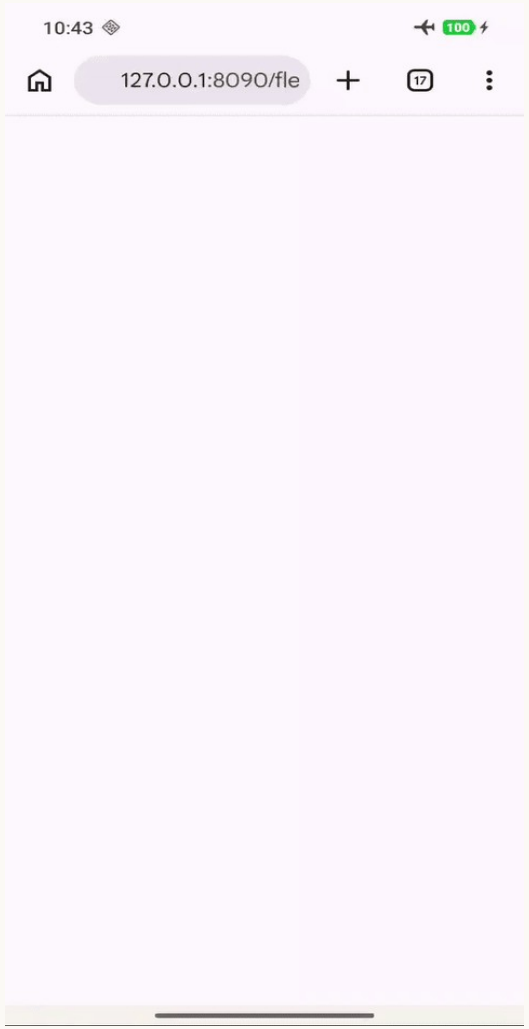
Fleets are our earliest revenue wedge. Health and environment come next on the same engine.

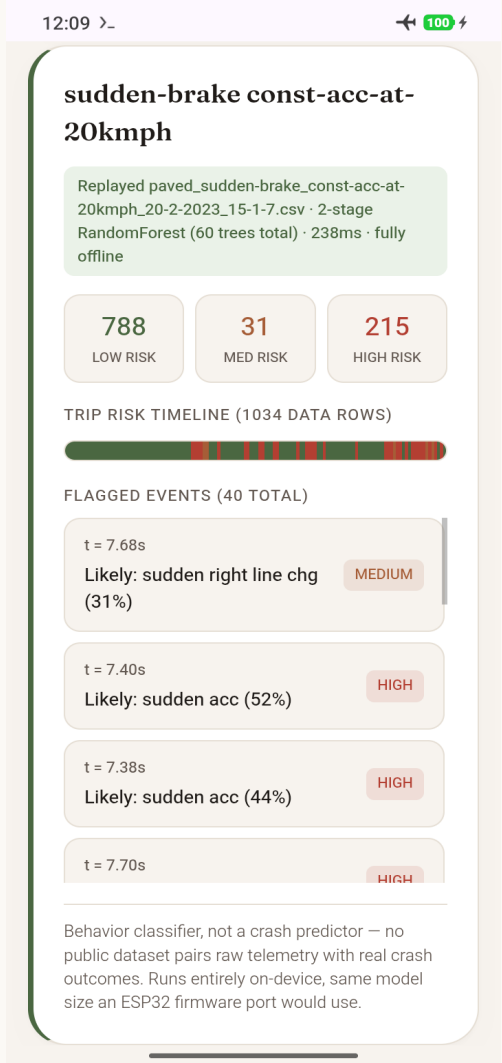
NOW

Edge AI fleet unit

- Crash detection
- EV battery health estimation
- Driver behaviour scoring
- Works offline, syncs when connected

Stage: Crash detection development and driver behaviour scoring is completed and battle testing with various datasets. Field testing collaboration is done with Local Private Transportation.





BENCHMARKS & PERFORMANCE

Measured, not estimated.

Tested on: **Redmi 24116RNC1I · Android 16 · 3.5GB RAM** (~1.6GB idle free). No laptop numbers used.

Model	Answer time	Quality
1.5B params Q4_K_M quantized	**42–84s**	Correct, grounded, no hallucination
0.5B params task-tuned layers	**15.3s** single **15–55s** load	Correct, grounded — zero quality loss

THERMALS

34.0°C → 35.0°C

Battery temp across ~25-call run

MEMORY FOOTPRINT

~30–50MB

RAM truly free with model loaded

AGRILENS

99.997%

Confidence on clean lab photo

FLEET RISK

54ms

Full 1,034-row trip replay

On-device execution guarantee: Zero cloud latency dependencies, full offline operation maintained across all edge intelligence workloads.

LIVE DEVICE HARDENING

Hardened before demo.

Adversarially tested on the live device

Confidence gating (AgriLens knows what it doesn't know)

A 70% confidence floor catches a low-confidence photo and asks for a retake instead of presenting a false diagnosis.

Grounding precision (LearnLoop never answers off memory)

Retrieval is filtered against common function words, so a shared word like "the" can't spuriously match an unrelated note.

Input safety (Fleet Risk only reads its own trip folder)

Trip selection is checked against a real directory listing before any file opens — closing a path-traversal route.

Open, tracked (LearnLoop's pace under a long session)

Answer time drifts from ~15s to 30–55s across repeated questions. Thermal throttling ruled out; tracked for a closer look.

DEPLOYMENT STATUS, HONESTLY

Where each piece actually stands

AgriLens

AgriLens field pilot

IN PROGRESS
Piloting with 5 farmers

ACTIVE DEPLOYMENT

5 Farmers

Live agricultural field testing

LearnLoop

LearnLoop field pilot

IN PROGRESS
Piloting with 10 students

ACTIVE DEPLOYMENT

10 Students

Classroom pilot evaluation

Fleet Risk

Database COMPLETE
On-device schema built, verified

Vehicle Hardware IN PROGRESS
Building the on-vehicle sensor layer

Cloud Compute IN PROGRESS
AWS aggregation layer being built

VISION & ROADMAP

A better India ahead: 100+ sector-specific models

100+ Models, On Any Smartphone

Once the architecture layers, training process, and performance benchmarks are finalized across sectors, the target is 100+ fully-trained, customized models — each tuned to its own sector the way AgriLens and LearnLoop already are.

All runnable on a smartphone of even the lowest practical configuration.

Where Latency Is Headed

These are unoptimized-pipeline numbers, not a ceiling

The model's own layers and routing between retrieval and generation are actively being tuned — newer quantization, decoding, and routing algorithms are being tested now.

TODAY

15–55s

per answer

TARGET, END OF 2026

A couple of seconds

per answer

Aligned with National Missions

- Startup India / DPIIT recognition
- IndiaAI Mission
- Digital Agriculture Mission
- PM eVidya / NEP 2020

VYOMAA LABS claims no formal affiliation with any of these programs beyond DPIIT recognition itself — **the aim is to build toward the same goals, for the same people.**

MARKET SCOPE

Hundreds of millions need AI but lack the internet and budget cloud AI demands.

958M

Active internet users in India; about 548M (57%) in rural areas.

Source: IAMAI, 2025

14.7 lakh

Schools; 63.5% have internet.

Source: UDISE+ 2024-25

14.6 crore

Operational farm holdings; 86% small or marginal.

Source: Agriculture Census 2015-16

7.4M

Commercial vehicles on Indian roads; about 3.5M with telematics.

Source: Brickwork, 2024

An active internet user does not always have a reliable connection in the field or classroom.

BUSINESS MODEL

Free or low-cost for people. Paid by institutions that already have budgets.

Schools and universities

Per-seat and per-institution licences for LearnLoop.

Agri co-ops and FPOs

Farmer advisory subscriptions and local crop-knowledge packs.

Commercial fleets

Edge hardware unit plus a per-vehicle monthly subscription.

Developers and OEMs

PocketMind SDK licence and domain model packs.

Governments

Deployment, localisation and citizen-service contracts.

Clinics (next)

MedNotes licence per practitioner.

ROADMAP AND ASK

Our mission: AI accessible to all. Next 12 months: from pilots to paying institutions.

12-month roadmap

0-3 months	Harden PocketMind on mid-range phones; launch the website
3-6 months	Pilots with a school, FPO or KVK and a fleet; first letter of intent; first government pilot application
6-9 months	First paying institutional customers; local-language packs
9-12 months	Expand to more districts and states; seed or grant funding

DPIIT ask

Startup India benefits, access to government procurement and grants, and IP support.

DEMO & RESOURCES

See it live: Everything in this deck, interactive

INTERACTIVE

Full Product Site

Explore full architecture breakdown, live on-device demo recordings, measured performance metrics, and in-depth reliability notes.

vyomaalabs.github.io/pocketmind

VIDEO

Full Demo Video

Watch core tools in action with a complete, narrated walkthrough detailing the underlying system architecture.

vyomaalabs.github.io/pocketmind/media/demo-full-narrated.mp4

VYOMAA LABS · AI in every hand, for a future-ready India.